

Yash Gupta

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Education

Bennett University

Bachelor of Technology in Computer Science and Engineering – CGPA: 9.2/10

Greater Noida, India

Expected Graduation: 2028

Technical Skills

Languages: C++, Python, JavaScript (ES6+), SQL, Bash

Frameworks & Backend: Node.js, Express.js, MERN Stack, FastAPI, WebSockets, RESTful APIs

Databases & Storage: PostgreSQL, MongoDB, SQLite, Query Optimization, Connection Pooling

Cloud & DevOps: AWS (EC2, S3, IAM, VPC), Terraform, Docker, GitHub Actions CI/CD, Linux CLI

Core Computer Science: Data Structures & Algorithms, Concurrency & Async I/O, OOP, System Design

Professional Experience

Probus

Noida, India

Software Development Engineer Intern

Jun 2026 – Aug 2026

- Engineered the core tariff calculation engine for enterprise platform **PROLT**, enabling real-time client valuation against live **IEX market** clearing rates.
- Architected normalized **PostgreSQL** schemas for high-frequency time-series datasets and implemented composite indexes, decreasing dashboard query latency by **40%**.
- Prevented Node.js event-loop blocking by designing asynchronous batch-processing pipelines with **pg-pool**, sustaining **p99 response latency under 200ms** under peak ingestion loads.
- Eliminated transaction discrepancies across financial settlement paths by integrating structured telemetry, granular error-handling middleware, and strict state assertions.

Xebia

Remote

Cloud & DevOps Engineering Trainee

May 2026 – Jun 2026

- Accelerated deployment velocity by **65%** through multi-stage CI/CD pipelines in **GitHub Actions**, automating code linting, automated validation gates, and artifact delivery.
- Provisioned reproducible staging baselines across **AWS EC2** and **S3** utilizing **Linux bash scripts** and AWS CLI automation, removing environment configuration drift.
- Hardened repository and deployment security by enforcing least-privilege IAM policies, automated branch-protection rules, and secure secret injection within self-hosted Linux runners.

Projects

Campus Desk – Real-Time Seat Reservation Engine | *React.js, Node.js, Express.js, MongoDB, WebSockets*

2026

- Engineered an interactive seat reservation platform utilizing bidirectional **WebSockets** to broadcast floor-plan availability, resolving concurrent double-booking anomalies.
- Designed optimized MongoDB document structures and compound indices, maintaining consistent sub-second state synchronization during peak campus booking intervals.
- Containerized application services using **Docker** and established automated testing checkpoints to validate multi-user reservation workflows before production deployment.

CampusX – Distributed Marketplace | *React.js, Node.js, Express.js, MongoDB, Socket.io, AWS (EC2, S3)*

2025

- Built a distributed peer-to-peer commerce platform featuring real-time messaging and direct media offloading to **AWS S3** via secure pre-signed URLs, cutting API server I/O overhead by **80%**.
- Optimized high-cardinality item searches by implementing indexed MongoDB schemas, achieving sub-50ms query responses under heavy catalog read traffic.
- Secured client endpoints using stateless **JWT authentication** and granular role-based access control (RBAC), deploying the backend on **AWS EC2** behind an Nginx reverse proxy.

Certifications

- AWS Certified Solutions Architect – Associate** (SAA-C03) | *Amazon Web Services*
- AWS Certified Cloud Practitioner** (CLF-C02) | *Amazon Web Services*

Achievements

- Hackathon Team Lead:** Led a team of 4 to **Rank 26 / 591 Teams (Top 4.4%)** at **BUSIH 2026**, defining system architecture, data models, and RESTful API specifications.
- Competitive Programming:** Solved **300+ algorithmic problems** on LeetCode and CodeChef, focusing on graph traversal, dynamic programming, and high-efficiency data structures.
- Distributed Systems:** Built resilient, low-latency backend architectures with a strong focus on high availability, database index tuning, and concurrent transaction safety.